DOES PATCH SAMPLING STRATEGY AFFECT U-NET SEGMENTATION PERFORMANCE ON PLANT ROOT IMAGES?

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1 INTRODUCTION

Plant root systems play a fundamental role in nutrient uptake, water absorption, and overall plant growth and development. Accurate quantification of root morphology — including root length, density, and branching patterns — is essential for plant phenotyping research, crop improvement, and stress tolerance studies (Smith et al., 2020; Han et al., 2023). However, obtaining these measurements manually is an extremely time-consuming and labour-intensive process, requiring trained agronomists to annotate individual root pixels across hundreds of images. This bottleneck has motivated a growing body of research into automated root segmentation methods using deep learning.

The U-Net architecture, introduced by Ronneberger et al. (2015), has become the dominant framework for biomedical and biological image segmentation due to its encoder-decoder structure with skip connections, which enables precise pixel-level localisation even when trained on very few annotated images. Smith et al. (2020) demonstrated that U-Net can successfully segment plant roots from rhizotron images, achieving an F1 score of 0.7 using only 50 annotated images. Subsequent studies have extended this work by proposing architectural improvements to U-Net, such as replacing the encoder with ResNet and incorporating attention mechanisms, achieving pixel accuracy above 99% on clean laboratory root images (Han et al., 2023). These results suggest that deep learning is a viable and increasingly powerful tool for automated root segmentation.

However, a critical and underexplored challenge remains: real-world root imaging conditions are far from clean. In petri dish imaging setups, condensation frequently forms on the dish surface, producing background noise, blurred boundaries, and uneven illumination that fundamentally alter the visual appearance of root structures. Under these conditions, the performance of standard U-Net trained with random patch sampling degrades significantly, as the model is not exposed to a sufficient number of challenging patches during training. Teramoto and Uga (2024) demonstrated that even moderate levels of noise can substantially reduce segmentation accuracy in root imaging, and explicitly noted that detailed studies on segmenting noisy root images remain scarce.

A key but largely overlooked factor in U-Net training is the patch sampling strategy — the method by which small image patches are selected from full-resolution images during training. Standard practice relies on random patch sampling, which produces a severe class imbalance: in root images, foreground root pixels represent only a small fraction of total image area, meaning the vast majority of randomly sampled patches contain only uninformative background (Smith et al., 2020; Tappeiner et al., 2022). Berger et al. (2017) demonstrated that adaptive sampling from high-error regions can dramatically accelerate training convergence, and Fischer et al. (2025) showed that patch sampling strategy introduces meaningful performance variability across segmentation tasks. Despite these findings, no study has investigated whether patch sampling strategy specifically affects U-Net performance on plant root images under condensation and background noise.

This study addresses this gap directly. The dataset used in this study consists of black and white morphometric images of *Arabidopsis thaliana* root systems, collected by the Hades phenotyping system at the Netherlands Plant Eco-phenotyping Centre (NPEC). Each petri dish was sown with five *A. thaliana* seeds and photographed daily as roots developed from germination through lateral root formation. Images were captured under controlled laboratory conditions using the Hades in-vitro root phenotyping installation, with visual variation across images arising primarily from condensation forming on the petri dish surface.

1.1 Research Question

The central research question of this study is:

Does patch sampling strategy affect U-Net segmentation performance on plant root images under condensation and background noise?

Three strategies are compared: random patch sampling as the baseline, weighted sampling that oversamples patches containing root pixels, and hard-example sampling that preferentially selects patches from high-noise condensation-affected regions. All experiments use an identical U-Net architecture and dataset, ensuring that any observed performance differences are attributable solely to the sampling strategy.

2 LITERATURE REVIEW

2.1 Foundational Architecture and Root Segmentation

The U-Net architecture, introduced by Ronneberger et al. (2015), established the foundational framework for biomedical image segmentation through its symmetric encoder-decoder structure with skip connections. The architecture captures both local fine-grained details and broader contextual information by combining a contracting path for feature extraction with an expansive path for precise pixel-level localisation. Trained end-to-end on as few as 20 to 35 annotated images using aggressive data augmentation and a weighted loss function to address class imbalance, U-Net achieved state-of-the-art performance across multiple biomedical segmentation benchmarks. However, all experiments were conducted on clean two-dimensional microscopy images, and patch sampling during training was random with no investigation into whether the composition of training patches affects model performance. Furthermore, the architecture was never evaluated under noise, condensation, or other real-world visual degradation conditions, leaving open the question of how robust U-Net is when imaging conditions are challenging.

Smith et al. (2020) demonstrated the feasibility of U-Net for plant root segmentation, training on only 50 annotated chicory root images from a rhizotron facility and achieving an F1 score of 0.7 on clean test images. Their training procedure extracted 90 random patches per image per epoch, filtered out patches containing no roots, and retained a maximum of 40 patches per image — a simple foreground filter rather than a systematic sampling strategy. Their dataset consisted exclusively of clean images with no condensation or background noise, and the only reported visual challenge was occasional scratches on the acrylic glass surface. Crucially, the authors themselves acknowledged that excluded background patches may contain hard negative examples that the network must learn to handle, and suggested that active learning could be used to identify difficult image regions — yet no experiments were conducted to address this limitation. This gap directly motivates the present study.

2.2 Root Segmentation Under Challenging Conditions

Han et al. (2023) proposed an improved U-Net architecture for plant root segmentation in minirhizotron images, replacing the standard encoder with ResNet50 and incorporating a Pyramid Split Attention module to improve segmentation of fine root edges. A combined Dice-Focal loss function was used to address the severe class imbalance, as 80% of images had roots occupying only a small portion of the image. The improved model achieved an F1-score of 95.10% and an IoU of 95.48% on clean peanut root images, outperforming the original U-Net. However, all experiments were conducted on images collected under controlled laboratory conditions with no condensation or background noise, and patch sampling remained random throughout training. The study demonstrates that architectural improvements can enhance segmentation performance, yet the effect of patch sampling strategy on model performance under challenging visual conditions such as condensation and noise remains unexamined.

Teramoto and Uga (2024) investigated the effect of image noise on deep learning-based semantic segmentation of rice root systems using X-ray CT scanning. The noise in their study was caused by rapid scanning speeds — shorter scan times of 33 and 66 seconds produced high levels of random noise, low resolution, and blurred root boundaries, making it difficult for the model to distinguish roots from background. To address this, they compared training directly on noisy volumes against applying conventional median filtering as a preprocessing step before training. Their results showed that scan times below 150 seconds produced volumes too noisy for effective model training, while adding conventional filtering improved training efficiency by approximately 10% across all noise conditions. However, even with filtering, the final model failed to predict lateral roots because they were absent from the training data, highlighting that the training data composition was never strategically adjusted to include hard or underrepresented examples. The authors themselves acknowledged that detailed conditional studies on segmenting noisy root segments remain scarce, directly confirming the research gap that the present study addresses.

2.3 Patch Sampling Strategy and Class Imbalance

Berger et al. (2017) proposed an adaptive patch sampling algorithm called isample for training fully convolutional networks on large and imbalanced medical image datasets. Rather than selecting patches randomly, their method generated error maps throughout training to identify regions where the model was performing poorly, and subsequently oversampled patches from these difficult regions. Applied to kidney segmentation from full-body CT scans — where the target class occupied only 0.3% of the total volume — isample achieved a Dice score of 0.855 after only 5,000 training iterations, compared to 40,000 iterations required by random sampling to reach similar performance, demonstrating that intelligent patch sampling from difficult regions can substantially improve both training efficiency and final model performance. However, their experiments were conducted exclusively on clean three-dimensional medical CT volumes, and the method was never evaluated on two-dimensional images affected by noise or condensation. The applicability of this adaptive hard-example sampling approach to plant root segmentation under noisy and condensation-affected conditions therefore remains an open research question.

Tappeiner et al. (2022) investigated the class imbalance problem in deep learning-based segmentation of head and neck organs, demonstrating that patch size is a direct hyperparameter controlling class imbalance during training — smaller patches increase the probability that underrepresented classes appear within each training patch, thereby improving learning for minority classes. The standard nnU-Net framework used a fixed 33% foreground oversampling strategy, but this was insufficient for highly sparse classes because when a class is absent from a patch, its contribution to the loss function becomes zero and the network receives no gradient signal for that class. To address this, the authors optimised patch size and introduced a class-adaptive Dice loss that excludes missing classes from loss computation, producing a 3% improvement in Dice score and a 22% reduction in Hausdorff distance without any architectural modifications. However, all experiments were conducted on clean three-dimensional CT volumes, and the patch sampling strategy itself was never investigated in the context of noisy or degraded images.

Fischer et al. (2025) demonstrated that patch sampling strategy is a critical and underexplored variable in medical image segmentation, proposing a Progressive Growing of Patch Size curriculum that systematically varies patch size throughout training. Their core insight was that smaller patches at the start of training produce stronger class balance, while larger patches later in training provide broader spatial context. Evaluated across 15 diverse 3D medical segmentation tasks, their method achieved a statistically significant mean Dice score improvement of 1.28% over the constant patch size baseline while reducing training time to 89% of the original, with benefits being most pronounced for class-imbalanced tasks. Critically, they also demonstrated that patch sampling strategy introduces meaningful stochastic variability — repeated training runs with different sampling strategies can yield substantially different models. However, all experiments were conducted on clean three-dimensional medical imaging datasets with no investigation into how sampling strategy interacts with images affected by noise or visual degradation.

2.4 Hard Example Mining

Shrivastava et al. (2016) introduced Online Hard Example Mining (OHEM), a foundational algorithm for addressing severe class imbalance in neural network training. Their core observation was that training datasets contain an overwhelming number of easy background examples relative to hard foreground examples, and that random sampling forces the network to waste most of its learning capacity on uninformative easy examples. OHEM addresses this by computing the loss for all candidate training examples in each mini-batch and selecting only the highest-loss examples for backpropagation, performed entirely online without interrupting the training process. Applied to object detection, OHEM improved mean average precision from 74.6% to 78.9% on PASCAL VOC 2007. However, all experiments were conducted on standard clean image benchmarks, and the method was never evaluated in the context of image segmentation under noisy or visually degraded conditions.

Schmidt-Mengin et al. (2022) conducted the most directly relevant comparison to the present study by pitting OHEM against fixed foreground oversampling for patch-wise 3D U-Net segmentation of new multiple sclerosis lesions — a task characterised by extreme class imbalance. Their results revealed a critical finding — fixed uniform oversampling consistently outperformed OHEM in terms of final Dice score, while OHEM training was approximately twice as slow per epoch. The authors concluded that while OHEM shows theoretical promise, fixed oversampling strategies remain competitive and practically more efficient for highly imbalanced segmentation tasks. However, all experiments were conducted on clean longitudinal MRI data, and no comparison was made across different levels of image quality. Whether the relative advantage of weighted oversampling versus hard-example mining changes under challenging imaging conditions such as condensation and background noise remains entirely uninvestigated.

2.5 Research Gap

Ciga et al. (2021) demonstrated that patch sampling strategy is not a trivial implementation detail but a critical design decision that directly affects model performance, showing that injecting negative patches into training reduced the false positive rate from 62.5% to 25% and achieved a 53% reduction in tumour extent error in whole slide image analysis. Taken together, the reviewed literature establishes that while U-Net has been successfully applied to plant root segmentation, all existing studies rely on random or simple foreground-filtered patch sampling and evaluate performance exclusively on clean images. No study has systematically compared random, weighted, and hard-example patch sampling strategies for U-Net trained on plant root images under condensation and background noise. This gap forms the central motivation for the present study.

3 METHODOLOGY

3.1 Research Design

This study employs a quantitative experimental design to isolate the effect of patch sampling strategy on U-Net segmentation performance. Three experimental conditions are compared: random patch sampling as the baseline, weighted patch sampling that oversamples patches containing root pixels, and hard-example patch sampling that preferentially selects patches from high-noise image regions. All other experimental variables — including model architecture, dataset, preprocessing pipeline, hyperparameters, and evaluation metrics — remain identical across all three conditions. This controlled design ensures that any observed differences in segmentation performance are attributable solely to the patch sampling strategy.

3.2 Dataset

The dataset consists of greyscale petri dish images of *Arabidopsis thaliana* root systems, collected by the Hades in-vitro phenotyping installation at the Netherlands Plant Eco-phenotyping Centre (NPEC). Each petri dish was sown with five seeds and photographed daily as roots developed from germination through lateral root formation. The dataset spans three academic year cohorts — referred to as Y2B_23, Y2B_24, and Y2B_25 — comprising 492 annotated image-mask pairs in total (Y2B_23: 115 images, Y2B_24: 314 images, Y2B_25: 63 images). All images are greyscale morphometric scans captured under controlled laboratory conditions, with variation in visual quality arising from condensation forming on the petri dish surface in a subset of images. Non-target images, including any potato dish images present in the raw data, were identified and excluded during preprocessing. Each image is accompanied by a binary root segmentation mask in which root pixels are labelled as foreground and all other pixels as background. Prior to training, all datasets are merged into a unified flat directory structure with consistent naming conventions. Images and masks are converted to greyscale PNG format and masks are binarised at a threshold of 127.

3.3 Exploratory Data Analysis and Baseline Results

Exploratory data analysis was conducted on the merged dataset of 492 annotated plant root images. Analysis of root pixel ratios revealed severe class imbalance, with a mean root pixel ratio of 0.71% and a median of 0.17%, indicating that the vast majority of pixels in each image represent background rather than root tissue. Twenty images (4.1%) contained no root pixels whatsoever. A patch sampling simulation demonstrated that under random patch sampling with a patch size of 256×256 pixels, only 14.5% of extracted patches contained at least one root pixel, meaning that 85.5% of randomly sampled patches consist entirely of background and provide no useful gradient signal for learning root features. A patch difficulty analysis on the training set (344 images, 17,200 patches sampled) further confirmed this finding: 85.5% of patches were completely empty, 0.7% contained trace root coverage (0–0.1% root pixels), 3.9% were sparse (0.1–1%), and only 9.9% contained clearly visible root coverage (>1%). These findings directly motivate the need for intelligent patch sampling strategies as proposed in this study.

To establish a performance baseline, a standard U-Net was trained using random patch sampling on the training split (344 images, 3,440 patches per epoch) with the combined Dice and binary cross-entropy loss function described in Section 3.9. Training employed the Adam optimiser with an initial learning rate of 0.001 and early stopping with a patience of 10 epochs, terminating at epoch 36 with a best validation Dice of 0.7892. The baseline model achieved a test Dice score of 0.7285 and a test IoU of 0.4005. The gap between validation Dice (0.7892) and test Dice (0.7285) suggests mild overfitting, consistent with the dominance of uninformative empty patches during training. The notably lower IoU (0.4005) relative to Dice reflects the model's tendency to produce false positive background predictions — a direct consequence of training predominantly on background-only patches.

3.4 Data Splitting

The merged dataset is partitioned into training, validation, and test subsets using a stratified split to ensure that images from all three academic year cohorts are represented proportionally across all subsets. The split ratio is 70% training (344 images), 15% validation (74 images), and 15% test (74 images). A fixed random seed of 42 is used throughout to ensure reproducibility. The test set is held out entirely during training and hyperparameter selection, and is used exclusively for final evaluation. Stratified splitting is chosen over random splitting because without stratification, images from a particular cohort could be disproportionately distributed across subsets, leading to biased evaluation results.

3.5 Preprocessing

All images are resized to a uniform resolution prior to patch extraction. Pixel values are normalised to the range [0, 1] by dividing by 255. Normalisation is applied because neural networks trained with gradient descent converge faster and more stably when input values are scaled, as large absolute values can cause gradient instability during backpropagation (Smith et al., 2020). No colour augmentation is applied as images are greyscale. Spatial augmentations including random horizontal and vertical flipping are applied to training patches only, using a shared random seed applied simultaneously to both image and mask to preserve spatial correspondence. Augmentation is not applied to validation or test patches to ensure unbiased evaluation, consistent with the approach of Ronneberger et al. (2015).

3.6 Patch Sampling Strategies

Three patch sampling strategies are implemented and evaluated. All strategies use a fixed patch size of 256×256 pixels, chosen as a balance between providing sufficient spatial context for the model to learn root structures and fitting within GPU memory constraints at batch size 8.

Strategy 1 — Random Sampling (Baseline): Patches are extracted from random locations within each training image at each epoch with no constraints on patch content. This strategy serves as the baseline because it reflects standard practice in the existing literature (Smith et al., 2020; Han et al., 2023).

Strategy 2 — Weighted Sampling (H1): At the start of each epoch, a foreground probability map is computed for each training image based on its root mask. Patch centres are sampled with probability proportional to the local root pixel density, ensuring that a greater proportion of training patches contain root pixels. This approach is motivated by the severe class imbalance observed in the EDA and supported by Tappeiner et al. (2022), who demonstrated that foreground-biased sampling improves segmentation performance for minority classes without any architectural changes.

Strategy 3 — Hard-Example Sampling (H2): An error map is maintained for each training image and updated at fixed intervals during training based on the pixelwise difference between the model's predictions and the ground truth masks. Patch centres are sampled with probability proportional to the local prediction error, forcing the model to focus on regions where it currently performs poorly. This strategy is motivated by the OHEM framework of Shrivastava et al. (2016) and validated in a segmentation context by Berger et al. (2017).

3.7 Hypotheses

To address the research question stated in Section 1.1, the following testable hypotheses are formulated:

H1: A weighted patch sampling strategy that oversamples patches containing root pixels will achieve a higher Dice score and IoU than random patch sampling on plant root images affected by condensation and background noise (Tappeiner et al., 2022; Smith et al., 2020).

H2: A hard-example patch sampling strategy that oversamples patches from high-noise image regions will outperform both random and weighted sampling on condensation-affected images, as measured by Dice score and IoU (Shrivastava et al., 2016; Berger et al., 2017).

3.8 Model Architecture

A standard U-Net architecture is used identically across all three experimental conditions (Ronneberger et al., 2015). U-Net is selected because it has been validated for plant root segmentation (Smith et al., 2020; Han et al., 2023), its encoder-decoder structure with skip connections enables precise pixel-level localisation of thin root structures, and it can be trained effectively from small datasets through data augmentation. Table 1 summarises the architectural details. Dropout regularisation at rate 0.2 is applied after each pooling operation to reduce overfitting on the limited training data.

Table 1. U-Net architecture details.

Component	Details
Encoder blocks	4 downsampling blocks
Conv layers per block	Two 3×3 convolutions
Normalisation	Batch normalisation + ReLU
Pooling	2×2 max pooling
Feature channels	64, 128, 256, 512
Decoder	Transposed convolutions + skip connections
Final layer	1×1 convolution + sigmoid
Dropout rate	0.2 after each pooling
Trainable parameters	31,042,369

3.9 Training Protocol

All three models are trained using identical hyperparameters to ensure fair comparison. The Adam optimiser is used with an initial learning rate of 0.001, selected over SGD because it adapts the learning rate per parameter and produces faster convergence on small imbalanced datasets. A ReduceLROnPlateau scheduler reduces the learning rate by a factor of 0.5 when validation loss plateaus for 5 consecutive epochs. Training proceeds for a maximum of 100 epochs with early stopping based on validation Dice with a patience of 10 epochs. A batch size of 8 is used. A combined loss function weights Dice loss at 0.8 and binary cross-entropy at 0.2, as Dice loss directly optimises the evaluation metric and handles class imbalance effectively (Smith et al., 2020). A fixed random seed of 42 is set for all random operations to ensure reproducibility.

3.10 Evaluation Metrics

Segmentation performance is evaluated using two complementary metrics: Dice Similarity Coefficient and Intersection over Union (IoU). Dice is selected as the primary metric because it directly measures overlap between predicted and ground truth segmentations and handles class imbalance effectively (Ronneberger et al., 2015; Smith et al., 2020). IoU is reported as a secondary metric because it penalises false positives more heavily than Dice, providing a complementary perspective on segmentation quality. Both metrics are computed on the held-out test set after training is complete.

3.11 Statistical Analysis

To determine whether observed differences in Dice scores between sampling strategies are statistically significant, the Wilcoxon signed-rank test is applied. This non-parametric test is chosen because it makes no distributional assumptions about the data, which is appropriate given the small test set of 74 images where normality cannot be reliably verified. Cohen's d effect size is reported alongside p-values to quantify the practical magnitude of performance differences. A significance threshold of $\alpha = 0.05$ is applied. Bootstrap resampling with 1,000 iterations is used to generate 95% confidence intervals for mean Dice differences between conditions.

4 Expected Outcomes

Based on the dataset properties identified in the exploratory data analysis and prior empirical findings in the literature, the following outcomes are anticipated. Table 2 summarises the predicted test performance for all three strategies relative to the baseline.

Table 2. Predicted test performance by patch sampling strategy.

Strategy	Expected Dice	Expected IoU	vs Baseline
Random (Baseline)	0.7285	0.4005	—
Weighted (H1)	0.76–0.80	Higher than baseline	+4–9pp
Hard-Example (H2)	0.78–0.83	Highest of three	+7–13pp

4.1 H1 — Weighted Sampling vs. Random Sampling

Weighted patch sampling is expected to achieve a higher Dice score and IoU than the random sampling baseline. Based on the EDA finding that 85.5% of randomly sampled patches are completely empty and only 14.5% contain any root pixels, weighted sampling is expected to substantially increase the proportion of informative training patches. This is expected to translate to a Dice score in the range of 0.76–0.80 (+4–9 percentage points above the baseline). The improvement in IoU is expected to be more pronounced than the improvement in Dice, as reducing background-only patches should directly reduce the false positive predictions observed in the baseline results. This prediction is supported by Tappeiner et al. (2022), who demonstrated that foreground-biased sampling produces measurable Dice improvements without any architectural modification. However, weighted sampling ensures only that patches contain roots — it does not prioritise the hardest or most ambiguous cases, meaning condensation-affected regions may remain underrepresented.

4.2 H2 — Hard-Example Sampling vs. Random and Weighted Sampling

Hard-example patch sampling is expected to achieve the highest Dice score and IoU of the three strategies, with a predicted Dice in the range of 0.78–0.83 (+7–13 percentage points above the baseline). By maintaining and updating a per-image error map throughout training and oversampling from high-error regions, the hard-example strategy is expected to systematically reduce prediction errors on condensation-affected and sparse-root images — the exact cases where the baseline model fails most frequently. This prediction is supported by Berger et al. (2017), who demonstrated that adaptive sampling from high-error regions reduces training iterations by a factor of eight compared to random sampling, and by Shrivastava et al. (2016), who showed that hard example mining substantially improves performance on difficult cases.

However, a critical caveat applies. Schmidt-Mengin et al. (2022) found that fixed foreground oversampling consistently outperformed OHEM for a highly imbalanced 3D segmentation task, and that OHEM training was approximately twice as slow per epoch due to error map computation overhead. If this finding generalises to the present two-dimensional setting, H2 may underperform H1 in overall Dice despite targeting harder examples. In this case, H1 would be supported but H2 would be rejected, providing evidence that weighted foreground oversampling is the more practical and effective strategy for plant root segmentation under the current dataset and training conditions.

Both hypotheses will be evaluated using the Wilcoxon signed-rank test with a significance threshold of $\alpha = 0.05$, Cohen's d effect size, and 95% bootstrap confidence intervals as described in Section 3.11.

5 TIMELINE

Table 3. Research timeline by week.

Week	Phase	Key Deliverable
1	Literature & Setup	Baseline confirmed
2	Weighted Sampling (H1)	H1 training complete
3	Hard-Example Sampling (H2)	H2 training complete
4	Analysis & Reporting	Report submitted

6 FEASIBILITY, ASSUMPTIONS, AND LIMITATIONS

6.1 Feasibility

The study is feasible within the available timeframe. The dataset is already collected and preprocessed, the baseline pipeline is fully implemented and validated, and both proposed strategies modify only the patch extraction step, keeping implementation effort limited. All experiments run on an NVIDIA RTX A6000 GPU with 48 GB VRAM.

6.2 Assumptions

- The 492 images are representative of petri dish *Arabidopsis thaliana* root images under condensation.
- A patch size of 256×256 pixels provides sufficient spatial context for learning root features.
- Pixelwise prediction error is a reliable proxy for patch difficulty in hard-example sampling.
- Baseline hyperparameters are equally appropriate for all three strategies.

6.3 Limitations

- Small test set (74 images) limits statistical power.
- Only a standard U-Net is tested — results may not generalise to other architectures.
- Single dataset domain — generalisation to other imaging setups cannot be guaranteed.
- Hard-example sampling introduces computational overhead that may disadvantage H2.
- No per-strategy hyperparameter tuning is performed.

7 ETHICAL AND DATA MANAGEMENT CONSIDERATIONS

7.1 Ethical Considerations

This study uses only greyscale images of *Arabidopsis thaliana* root systems. No human participants, personal data, or sensitive information are involved. Ethical approval is not required.

7.2 Data Management

- All data is stored on the BUas GPU server at `/home/y2b/`. No data is stored on personal devices.
- The original raw dataset is never modified — preprocessing produces copies only.
- A fixed random seed of 42 ensures full reproducibility.
- Model checkpoints are saved at each validation improvement. Only the best checkpoint per experiment is retained.
- Data is owned by BUas and will not be shared with third parties.

References

Berger, J.-O., Castellan, G., Niaf, E., and Lartizien, C. (2017). isample: A patch sampling algorithm for deep learning-based medical image segmentation. In *Proceedings of the IEEE International Symposium on Biomedical Imaging (ISBI)*, pages 680–683.

Ciga, O., Xu, T., and Martel, A. L. (2021). Self supervised contrastive learning for digital histopathology. *Machine Learning with Applications*, 6:100198.

Fischer, P., Zimmerer, D., Küstner, T., and Maier-Hein, K. (2025). Progressive growing of patch size for medical image segmentation. *Medical Image Analysis*, 101:103420.

Han, X., Zheng, H., and Chen, M. (2023). Improved U-Net for plant root segmentation in minirhizotron images. *Frontiers in Plant Science*, 14:1143731.

Ronneberger, O., Fischer, P., and Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, pages 234–241. Springer.

Schmidt-Mengin, M., Jobidon, V., Doré, F., Bhattacharya, P., Arbel, T., and Leppert, I. R. (2022). Online hard example mining versus fixed foreground oversampling for 3D U-Net segmentation of new MS lesions. arXiv preprint arXiv:2205.01626.

Shrivastava, A., Gupta, A., and Girshick, R. (2016). Training region-based object detectors with online hard example mining. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 761–769.

Smith, A. G., Petersen, J., Selvan, R., and Rasmussen, C. R. (2020). Segmentation of roots in soil with U-Net. *Plant Methods*, 16(1):13.

Tappeiner, E., Welk, A., Hänsch, R., and Menze, B. (2022). Class-adaptive loss weighting and patch sampling for imbalanced medical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2022*, pages 311–320. Springer.

Teramoto, S. and Uga, Y. (2024). Deep learning-based segmentation of rice root systems from X-ray CT images under varying noise conditions. *Plant Methods*, 20(1):45.